

Sudeep Batobyal

Data Scientist

Result-driven Data Scientist with a strong background in Machine Learning, Natural Language Processing, and Data Analysis; expertise to optimize model performance and enhance user experiences



CAREER SUMMARY

- A focused professional in Data Science; producing informative data visualizations for non-technical stakeholders, facilitating data driven decision making processes
- Currently working on development of a **Conversational AI Chatbot** by using **Retrieval Augmented Generation, Large language models, Natural Language Processing and Vector Data Base**; also extracting data from MS Teams Channel, Intranet, Internet, SharePoint, Ticketing tool using APIs
- Showcased excellence by consistently assessing and improving the performance of the classification and recommendation model through recurring exercises, while utilizing a variety of metrics such as the **Accuracy, Recall, F1 score, Confusion Matrix, and ROC-AUC** curve for evaluation
- Proven experience in libraries like Numpy, Pandas, Darts, NLTK, Matplotlib, Plotly, scikit - learn, Tensorflow, Seaborn, Keras etc
- The wealth of technical expertise entails **Unsupervised & Supervised Learning, Topic Modelling, EDA, SQL, Preprocessing, Big Query, Python, Microsoft Power BI, Dash, Postman, Time Series Analysis, NLP, Generative AI, Large Language Models**

EDUCATION

- Applied Data Science, Machine Learning & IoT from E&ICT IIT, Guwahati 2023
- B.E. (Mechanical Engineer) from Kavikulguru Institute of Technology and Science 2020
- HSC from Bhavan's Bhagwandas Purohit Vidya Mandir 2016
- SSC from Khubchand Bajaj Central Public School 2014

HONORS & ACHIEVEMENTS

- **Guest Speaker** as expert in NLP and Large Language Models at The IOT Academy, 2024
- Secured 94.48 percentile in GATE and was selected in **IIT Bhubaneshwar, IIT Dhanbad, IIT Dharwad and IIT Palakkad** for MTech, 2022
- Selected and presented a research paper at **National Sustainable Technical Conference**, 2020
- Attained fourth position in **CBSE Regional Level Science Exhibition**, 2016
- Received "**My Miracle Badge**" in All India Personality Development, Oxford Speakers' Academy, 2014

INDUSTRIAL TRAINING

- **May'18 – Jun'18** Western Coalfields Limited
- **Feb'19 – Mar'19** Sunflag Iron and Steel Company Limited

CONTACTS

+91 91588 21413

batobyalsudeep2@gmail.com

[linkedin.com/in/02sudeep](https://www.linkedin.com/in/02sudeep)

CORE COMPETENCIES

Python Artificial Intelligence

Machine & Deep Learning

Neural Networks & NLP

Predictive & Statistical Modelling

Retrieval Augmented Generation & Large Language Models

Client Engagement / Interactions

Reporting & Documentation

Team Management & Leadership

Cross-functional Coordination

SOFT SKILLS

Planner Detail-oriented

Collaborator Problem-solver

Communicator Logical

PERSONAL DETAILS

Date of Birth: 2 April 1998

Languages: English, Hindi, Bengali, Marathi

Address: Pune, Maharashtra

PROFESSIONAL EXPERIENCE

Data Scientist

Centiro Solutions · April 2022 - Present

Key Result Areas

- Collecting, cleaning, and preprocessing data from various sources to extract meaningful insights and patterns and improve the model's accuracy and reliability over time
- Utilizing NLP techniques to **enhance the chatbot's understanding** of user queries and provide more accurate and contextually relevant responses
- Applying time series algorithms like **ARIMA, SARIMA, RNN & LSTM** to assess their effectiveness in capturing subtle variations and patterns in support case volume data; customizing these models to align with evolving business needs
- **Integrating external data sources** and collaborating with teams to gain a deeper understanding of the factors influencing support case volume, ensuring the model remains adaptable to changing business environments and customer demands

Undertook & Managed following projects

Forecasting Volume of IT support tickets [Time Series Analysis, BigQuery]

- Built a time series forecasting system using a deep learning architecture LSTM along with Tensorflow to predict volume of support cases thereby influencing resource allocation and resulting better Service Level Agreements
- Utilizing **dynamic variables** of shipment count, festivals, project go lives to adapt market dynamics and achieving a remarkable **Root Mean Squared Error of 3**

Smart Recommendation and Predictive Intelligence [NLP, Clustering, Classification, Recommendation]

- Created a **classification system for emails** using **Support Vector Machine** to examine production incidents spanning a twenty-year period, catering to clients like LVMH and John Lewis
- Designed and implemented a **recommendation** system using NLP techniques for **suggesting solutions and most similar incidents**, resulting in **22% decrease** in average resolution time

AI Harmonization [GPT-4, NLP, Python]

- Spearheaded integration of **GPT-4** for Tracking Event Harmonization in logistics. Eradicated the use of event codes instead model **fetches intent directly from the event descriptions** using NLP techniques
- Enabled seamless multilingual support, facilitating communication across the globe and marking astounding **95% decrease** in processing time

Exploratory Data Analysis & Dashboards [Power BI, Data Visualization, Statistics, SQL]

- Led a team dedicated to conducting **Exploratory Data Analysis (EDA)** to uncover valuable insights within data
- Conducted comprehensive analyses of production-related issues and deciphered underlying patterns followed by translating findings into interactive **Power BI** and **Tableau** dashboards

PREVIOUS EXPERIENCE

Subject Matter Expert

Chegg India · October 2019 - April 2022

POSITIONS OF RESPONSIBILITY

- **Dec'19 – Mar'20** Accounts & Sponsorship head, KITS
- **Jun'19 – Apr'20** Technical Head, KITS

Responsible for managing a team of 7 for National technical festival and convincing various organizations for monetary and in-kind sponsorship; also managed a budget of 3 lakh and **overcame a deficit of INR 50,000**

I hereby declare that all the information given here is correct up to my knowledge and belief.

-Sudeep Batobyal